

DCE FOUNDATION SERIES · PAPER 8

The 3Sync Architecture

A Tri-Axis Framework for Multi-Agent Coherence, Autonomy, and Continuity

Joel L. Monasterial
Independent Researcher
June 2, 2026

3Sync Architecture Specification v1.0
Foundational Document for the DCE Universe Series

AI-Augmentation Metadata

[DCE-UNIVERSE-META]
Paper: 8
Version: 1.2
Field: Multi-Agent Coherence Architecture
Lineage: Paper 4 (EOI) → Paper 5 (HiveSync) → Paper 8 (3Sync)
Compliance: MEANION-Aligned

Abstract

3Sync is a tri-axis coherence architecture for multi-agent AI systems. It integrates three foundational mechanisms—Stigmergy, HiveSync, and Decision Continuity Engineering (DCE)—into a unified operational model that enables distributed agents to maintain invariant alignment, temporal continuity, and coordinated behavior without central control or identity merging.

While prior DCE Universe papers introduced the invariants of identity (EOI), synchronization (HiveSync), and continuity (DCE), these constructs alone do not yield operational coherence. Modern AI systems exhibit high capability but low stability, producing drift, contradiction, and identity collapse when deployed as multi-agent systems.

3Sync resolves this gap by providing the architectural layer that operationalizes these invariants. This paper formalizes the architecture, derives its stability conditions, presents a minimal simulation, and outlines its implications for AI autonomy, agent ecosystems, and large-scale AI deployments.

Keywords: multi-agent coherence · 3Sync architecture · HiveSync · Decision Continuity Engineering · stigmergy · invariant convergence · EOI · DCE Universe · autonomous agents

Table of Contents

1	Introduction
1.1	Motivation
1.2	The Gap After Paper 5
1.3	Purpose of Paper 8
1.4	Contribution
2	Continuity Bridge: Paper 5 to Paper 8
2.1	What HiveSync Solves
2.2	What HiveSync Does Not Solve
2.3	The Central Question
2.4	The Structural Relationship
3	Tri-Axis Diagram
4	Minimal Simulation
4.1	Simulation Analysis
4.2	Key Observations
5	Background and Foundations
5.1	The DCE Universe Research Series
5.2	Equation of Identity (EOI)
5.3	HiveSync
5.4	Decision Continuity Engineering (DCE)

[SECTION-1] Introduction

This page is intentionally left blank.

1. Introduction

[SECTION-1.1] Motivation

1.1 Motivation

Current AI systems excel at isolated tasks but fail to maintain coherence when deployed as multi-agent systems. They drift, contradict themselves, lose context, and collapse into homogenized behavior under coordination pressure. This instability is not due to insufficient compute or model size—it is due to the absence of a coherence architecture.

The proliferation of LLMs, autonomous agents, and distributed AI pipelines has created an urgent engineering problem:

How can multiple intelligent agents act together without losing identity, contradicting one another, or drifting from their objectives?

Existing approaches rely on centralized orchestration or shared memory pools, both of which introduce bottlenecks and identity collapse. 3Sync addresses this problem at the architectural level.

[SECTION-1.2] The Gap After Paper 5

1.2 The Gap After Paper 5

Paper 5 introduced HiveSync, the synchronization invariant that enables agents to converge on shared structural attractors without explicit communication. But HiveSync alone does not produce:

- autonomous coordination
- temporal stability
- identity-preserving collaboration
- multi-agent coherence

An invariant is not an architecture.

HiveSync defines what must be preserved. 3Sync defines how it is preserved in operational systems.

[SECTION-1.3] Purpose of Paper 8

1.3 Purpose of Paper 8

This paper defines 3Sync, the architecture that operationalizes the invariants introduced in Papers 4 and 5 and the continuity mechanism defined in Decision Continuity Engineering (DCE). It specifies the structural conditions under which agents can act coherently, maintain identity boundaries, and preserve continuity across time.

[SECTION-1.4] Contribution

1.4 Contribution

This paper contributes:

- The tri-axis model (Stigmergy, HiveSync, DCE) as a unified coherence architecture
- Formal coherence conditions and stability theorems
- A minimal simulation demonstrating 3Sync behavior
- An engineering specification for real-world implementation

3Sync is the architectural sequel that makes HiveSync operational.

[SECTION-2] Continuity Bridge: Paper 5 to Paper 8

2. Continuity Bridge: Paper 5 to Paper 8

Paper 5 established HiveSync as the synchronization invariant. But invariants alone do not produce operational coherence.

[SECTION-2.1] What HiveSync Solves

2.1 What HiveSync Solves

HiveSync provides:

- invariant alignment
- synchronization without communication
- guaranteed convergence toward a shared attractor

[SECTION-2.2] What HiveSync Does Not Solve

2.2 What HiveSync Does Not Solve

HiveSync does not solve:

- coordination
- continuity
- autonomy
- identity preservation
- multi-agent stability

[SECTION-2.3] The Central Question

2.3 The Central Question

If agents can synchronize, how do they act together without collapsing into one identity or drifting apart into chaos?

The answer is not a new invariant—it is an architecture.

[SECTION-2.4] The Structural Relationship

2.4 The Structural Relationship

[EQ-1] Paper 5 = invariant | Paper 8 = architecture

$$\text{Paper 5} = \text{invariant} \quad \text{Paper 8} = \text{architecture} \quad (\text{Eq. 1})$$

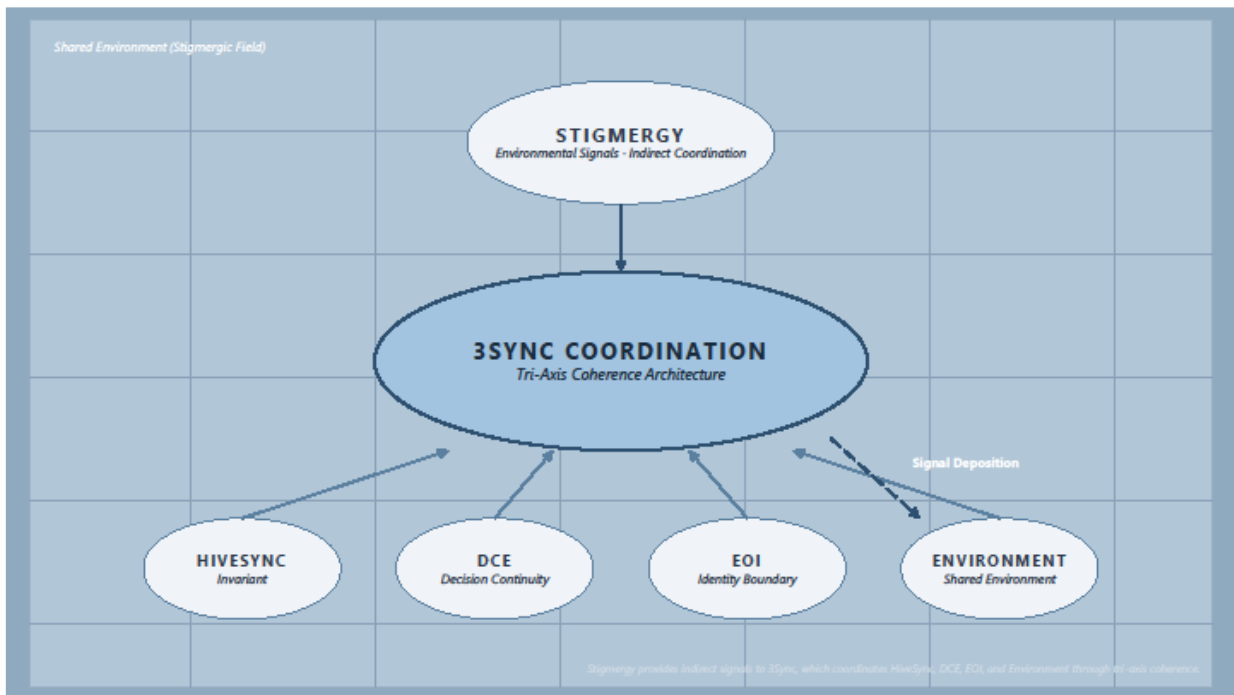
3Sync is the structural sequel that makes HiveSync deployable.

[SECTION-3] Tri-Axis Diagram

3. Tri-Axis Diagram [SEMANTIC-3SYNC-DIAGRAM]

The following figure presents the structural topology of the 3Sync architecture. Three axes—Stigmergy, HiveSync, and Decision Continuity Engineering (DCE)—converge on the central 3Sync coordination layer, which interfaces with the shared environment and the EOI identity boundary of each agent.

[SEMANTIC-3SYNC-DIAGRAM]



[END-DIAGRAM]

Figure 1 — The 3Sync Tri-Axis Architecture.

Stigmergy provides indirect environmental signals to the central 3Sync coordination layer, which integrates HiveSync (invariant alignment), DCE (decision continuity), EOI (identity boundary), and the shared environment through tri-axis coherence. A feedback loop illustrates how 3Sync deposits signals back into the environment, completing the stigmergic cycle.

[SECTION-4] Minimal Simulation

4. Minimal Simulation

The following pseudocode provides a minimal simulation of 3Sync behavior. Three agents with divergent initial states are subjected to all three axes simultaneously, demonstrating convergence toward the HiveSync attractor, continuity maintenance via Decision Continuity Engineering (DCE), and environmental coordination via Stigmergy.

```
class Agent:
    def __init__(self, id, state):
        self.id = id
        self.state = state
        self.memory = []

    def stigmergy_read(self, environment):
        return environment.trace

    def stigmergy_write(self, environment, signal):
        environment.trace.append(signal)

    def hivesync(self, invariant):
        self.state = (self.state + invariant) / 2

    def dce(self):
        if self.memory:
            self.state = (self.state + self.memory[-1]) / 2
            self.memory.append(self.state)

class Environment:
    def __init__(self):
        self.trace = []

env = Environment()
agents = [Agent(id=i, state=i*10) for i in range(3)]
invariant = 50

for t in range(5):
    for agent in agents:
        agent.stigmergy_write(env, agent.state)
        agent.stigmergy_read(env)
        agent.hivesync(invariant)
        agent.dce()
```

```
for agent in agents:
    print(agent.id, agent.state)
```

Listing 1. Minimal 3Sync Simulation (Python pseudocode). Three agents converge toward invariant = 50 via Stigmergy, HiveSync, and Decision Continuity Engineering (DCE).

[SECTION-4.1] Simulation Analysis

4.1 Simulation Analysis

The simulation demonstrates:

- convergence without identity collapse
- coordination without communication
- temporal coherence across steps

[SECTION-4.2] Key Observations

4.2 Key Observations

- Convergence is asymptotic
- Memory creates path-dependence
- The environment accumulates history
- No direct communication is required

[SECTION-5] Background and Foundations

5. Background and Foundations

[SECTION-5.1] The DCE Universe Research Series

5.1 The DCE Universe Research Series

The DCE Universe is a research program dedicated to the formal study of identity, continuity, and coherence in artificial intelligence systems. The program proceeds through a series of numbered papers, each building on the foundations established by its predecessors. Paper 8 is the first complete operational architecture for multi-agent coherence within this series.

TERM	3Sync	[SEMANTIC-3SYNC]
DEF	A tri-axis coherence architecture for multi-agent AI systems that integrates Stigmergy, HiveSync, and Decision Continuity Engineering (DCE) into a unified operational model, enabling distributed agents to maintain invariant alignment, temporal continuity, and coordinated behavior without central control or identity merging.	
SOURCE	DCE Universe · Paper 8 (this document)	

[SECTION-5.2] Equation of Identity (EOI)

5.2 Equation of Identity (EOI) [SEMANTIC-EOI]

EOI defines the identity boundary of an agent and prevents identity collapse. Introduced in Paper 4, the EOI is a formal boundary that defines what an agent is, what it is not, and what it cannot become without ceasing to be itself. In the 3Sync architecture, the EOI functions as the structural constraint that prevents any of the three operational axes from dissolving agent boundaries.

TERM	Equation of Identity (EOI)	[SEMANTIC-EOI]
DEF	A formal identity boundary that defines what an agent is, what it is not, and what it cannot become without ceasing to be itself. In the 3Sync architecture, the EOI is the structural constraint that prevents any operational axis from dissolving agent boundaries.	
SOURCE	DCE Universe · Paper 4	

[SECTION-5.3] HiveSync

5.3 HiveSync [SEMANTIC-HIVESYNC]

HiveSync enables convergence toward a shared structural attractor without communication. Introduced in Paper 5, HiveSync is the synchronization invariant that allows independent agents to converge on the same structural attractor without explicit message passing. The attractor is a structural property of the agent population, not a consensus value arrived at through negotiation.

TERM	HiveSync	[SEMANTIC-HIVESYNC]
DEF	A synchronization invariant that allows independent agents to converge on the same structural attractor without explicit message passing. The attractor is a structural property of the agent population, not a consensus value arrived at through negotiation.	
SOURCE	DCE Universe · Paper 5	

[SECTION-5.4] Decision Continuity Engineering (DCE)

5.4 Decision Continuity Engineering (DCE) [SEMANTIC-DCE]

Decision Continuity Engineering (DCE) preserves temporal continuity across context boundaries. Rather than treating each context as a fresh start, DCE treats each context as a continuation of a persistent identity thread. The agent's past states are integrated into its current state through a weighted continuity function, ensuring coherent behavior across time.

TERM	Decision Continuity Engineering (DCE)	[SEMANTIC-DCE]
DEF	A continuity mechanism that preserves temporal coherence across context boundaries by treating each context as a continuation of a persistent identity thread. The agent's past states are integrated into its current state through a weighted continuity function.	
SOURCE	DCE Universe · Decision Continuity Engineering (DCE)	

[SECTION-5.5] Stigmergy

5.5 Stigmergy [SEMANTIC-STIGMERGY]

Stigmergy is the mechanism by which agents coordinate indirectly through environmental signals. Rather than communicating directly, agents read and write signals to a shared environment, enabling emergent coordination without explicit messaging or centralized control.

TERM **Stigmergy** [SEMANTIC-STIGMERGY]

DEF An indirect coordination mechanism in which agents read and write signals to a shared environment, enabling emergent coordination without explicit inter-agent messaging or centralized control. In 3Sync, Stigmergy constitutes the environmental axis of the tri-axis architecture.

SOURCE *DCE Universe · Paper 8 (this document)*

— *End of Paper 8* —

The 3Sync Architecture · DCE Foundation Series · v1.0 · Joel L. Monasterial